

Keegan Li

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EDUCATION

Columbia University

Sept. 2021 - May 2025

B.S. in Computer Science

New York, New York

- Dean's List. Lawrence Gussman Scholarship recipient. Co-founder of Columbia Science Olympiad. New Student Orientation Program Leader, NCAA Division 1 Track & Field Athlete (Team Captain), Columbia Poker Club, Columbia Chess Club, Columbia Asian & Pacific Islander Student-Athlete Association. **Relevant Coursework:** Natural Language Processing, Artificial Intelligence, Machine Learning, Databases, Computer Networks, UX/UI Design, Cloud Computing, Analysis and Optimization, Data Structures in Java, Advanced Programming in C, Probability and Statistics, Linear Algebra.

EXPERIENCE

AI/ML Research Assistant

Jan. 2025 – Present

Columbia University

New York, New York

- Currently a research assistant to Professor Junfeng Yang at Columbia University, investigating causal mechanisms of large language model hallucinations and developing systematic mitigation methodologies.
- Investigating causal mechanisms behind large language model hallucinations through systematic prompt engineering and controlled experiments, developing methodologies to trace and categorize different types of factual inconsistencies.
- Implementing diagnostic frameworks to analyze attention patterns and hidden state activations during hallucination events, mapping the relationship between model architecture and false information generation.
- Developing quantitative metrics for hallucination detection and classification, leveraging statistical analysis to identify correlations between model behavior and output reliability.
- Skills:** Data Cleaning, Model Evaluation Metrics, Neural Network Debugging, Transformer Architecture, Attention Mechanism Analysis, Statistical Hypothesis Testing, PyTorch, TensorFlow, HuggingFace, LLM Fine-tuning, Python

LLM Research Assistant

Sept. 2024 – Present

Columbia University

New York, New York

- Currently a research assistant to Professor Venkatasubramanian in the CRIS Lab working on geometric information processing in LLMs. Implementing sparse autoencoders to analyze n-gram probability distributions across GPT-2 Small and TinyBERT, providing insights into how different model architectures encode semantic information.
- Developed novel approaches to model interpretability by tracking neuron activation patterns across diverse prompts and text genres using sparse autoencoders, revealing distinct geometric patterns in how LLMs process different types of content.
- Replicated Anthropic's monosemanticity experiments with smaller open-source LLMs, conducting dictionary learning and neural clamping experiments to map the geometric representation of semantic knowledge within model architectures.
- Skills:** Retrieval-Augmented Generation, Fine-tuning, LLM Training, Pandas, NumPy, Scikit-learn, Data Processing

Machine Learning Engineer Intern

May 2024 – Aug. 2024

Muse Tax

New York, New York

- Worked at a fintech startup **finetuning** Meta LLaMa 3 and Anthropic Claude models with **AWS SageMaker (feature extraction methodologies)** to develop an AI application that generated user-personalized financial advice.
- Implemented Retrieval-Augmented Generation (RAG) using **AWS BedRock** and **AWS OpenSearch** to enhance chatbot responses. Integrated IRS data into the retrieval process to provide more accurate and relevant financial advice.
- Developed IRS data scraping and vectorization pipelines using the **Sentence Transformers** library.
- Applied **prompt engineering** methodologies from OpenAI and Anthropic documentation to tailor model behavior.
- Created a CI/CD pipeline using **AWS Lambda** to process text from IRS manuals, enhancing data extraction/processing.
- Skills:** Linux, AWS SageMaker, AWS S3, AWS Lambda, Amazon API Gateway, AWS DynamoDB, AWS Bedrock, AWS Knowledge Bases, Aurora, OpenSearch, Natural Language Processing, Machine Learning, JSON, Python

Software Engineer Intern

Jan. 2023 – Apr. 2024

AI Media Technologies

New York, New York

- Developed automated testing algorithms for **RESTful API testing** with **Postman** and **smoke testing** techniques.
- Analyzed historical data stored in **AWS DynamoDB** and created data pipelines with **AWS Lambda**.
- Authored **API and system documentation** while maintaining backend for compatibility with monthly product updates.
- Created "nanny algorithms" to monitor database queries, eliminating need for human monitoring.
- Performed successful **migration of database** from AWS DynamoDB to **AWS S3** using AWS Lambda and **AWS Glue**.
- Skills:** Python, JavaScript, HTML, C#, C++, AWS DynamoDB, AWS Lambda, AWS S3, AWS Glue, Git, HTTP Requests, Web Server Programming, Smoke Testing

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C, C#, HTML, JSON, AWS CLI/SDK, SQL, TypeScript, C++

Techniques: Machine Learning, AWS architecture design and deployment, RAG, Fine-tuning, Prompt Engineering (Anthropic, OpenAI, Meta LLaMa), Natural Language Processing, Feature Extraction, ML Modeling, A/B Testing Models, Smoke Testing, RESTful API Testing, Web Scraping, Data Cleaning, Data Structures, Algorithms, Kubernetes, Microservices

Databases: MongoDB, AWS S3, AWS OpenSearch, AWS DynamoDB, AWS RDS, Pinecone, AWS Aurora, MySQL

Tools: GitHub, Linux, Node.js, TailwindCSS, React, Flask, AWS SageMaker, AWS S3, AWS Lambda, AWS API Gateway, AWS DynamoDB, AWS BedRock, AWS Knowledge Bases, AWS Q, Containerization (EC2, ECS, EKS), AWS Kendra, AWS Guardrails

Libraries: Pandas, NumPy, Matplotlib, Scikit-learn, Sentence-Transformers, Boto3, Unittest, PyTest, Socket